SCHOLAR — FINAL MASTER DEVELOPMENT PROMPT

Build a production-ready internal web application called **Scholar — Internal Research Operating System**.

This application is an internal research, scholarship, university, professor, opportunity, and content-management platform.

The application must be database-driven, scalable, secure, searchable, relational, and maintainable.

This is NOT a static website, NOT a simple spreadsheet mockup, and NOT a collection of disconnected pages.

Use the existing project/repository if one already exists. Before coding, inspect the existing codebase and preserve all working functionality. Reuse existing components, utilities, authentication, database schemas, and UI patterns whenever appropriate.

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1. CORE TECHNOLOGY

Use:

- React
- TypeScript
- Tailwind CSS
- Supabase
- PostgreSQL
- Supabase Authentication
- Row Level Security (RLS)

Use reusable components and a centralized design system.

The database must be the source of truth.

Do not hard-code records.

Do not create artificial limits such as:

- maximum countries
- maximum universities
- maximum departments
- maximum professors
- maximum scholarships
- maximum content
- maximum opportunities

The system must scale to thousands or tens of thousands of records.

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2. AUTHENTICATION

This is an internal application.

There must be NO public registration.

Only authorized users can access the system.

Authentication flow:

1. Admin invites a user by email.
2. User receives invitation email.
3. User verifies/authenticates through email.
4. User gets access to the application according to their role.
5. Protected routes prevent unauthorized access.
6. Logout must work.
7. Secure session management must be implemented.

Never use:

- shared passwords
- hard-coded passwords
- plaintext passwords
- public registration

Use Supabase Auth securely.

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3. USER / TEAM SYSTEM

Create a reusable `profiles` system.

Fields:

- id
- full_name
- email
- avatar_url
- role
- active
- created_at
- updated_at

Roles can include:

- Admin
- Researcher
- Verifier
- Content Manager
- Editor
- Designer
- Viewer

Roles and permissions must be configurable.

Use the same user/profile system throughout the application.

Examples:

- Assigned Researcher
- Assigned Verifier
- Assigned Content Manager
- Assigned Designer
- Assigned Editor
- Updated By
- Verified By

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4. MAIN APPLICATION STRUCTURE

After login, show a professional dashboard with sidebar navigation.

Main sections:

1. Dashboard
2. Scholarships
3. Universities
4. Professors
5. Co-Curricular
6. Content
7. Deadlines
8. Verification
9. Team
10. Activity Log
11. Settings

The UI should feel like:

- Notion
- Airtable
- Google Sheets
- modern research/admin software

But create an original Scholar visual identity.

Design:

- clean
- professional
- research-oriented
- minimal
- fast
- desktop-first
- responsive
- excellent spacing
- clear typography
- subtle borders
- minimal shadows
- consistent badges
- no unnecessary animation

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5. IMPORTANT ARCHITECTURE RULE

Do NOT unnecessarily connect unrelated modules.

There are three important architectural distinctions:

A. PROFESSOR SECTION IS INDEPENDENT

The Professor section must NOT depend on the University module.

The Professor module has its OWN hierarchy:

Country
→ University
→ Department
→ Professor

This university is only a professor-research organizational record.

It is NOT the same database relationship as:

Universities
→ Programs
→ Admissions
→ Intakes
→ Tuition

The Professor module must work independently.

A university can exist in the Professor section even if it does not exist in the main University section.

B. CO-CURRICULAR IS AN INDEPENDENT OPPORTUNITY MODULE

Co-Curricular contains exactly four main subsections:

1. Leadership
2. Fellowship
3. Internship
4. Emergency Upload / Recent News Update

Do not add unnecessary categories such as Postdoctoral, Volunteering, Research Opportunity unless explicitly requested later.

C. CONTENT IS THE CENTRAL CONTENT-MANAGEMENT WORKSPACE

The Content section connects to:

- Scholarships
- Co-Curricular
- Deadlines
- Application-start information
- Newly uploaded information
- Emergency Upload / Recent News Update

The Content Manager should not have to open multiple modules repeatedly just to understand what content needs to be published.

The Content Dashboard should aggregate all relevant information.

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6. DASHBOARD

Create an overall dashboard.

Show:

Main statistics

- Total Scholarships
- Total Universities
- Total Professors
- Total Countries
- Total Co-Curricular Opportunities
- Total Content
- Published Content
- Ready to Publish
- Upcoming Deadlines
- Emergency Uploads

Deadline summary

Show:

- Next 5 days

- Next 7 days
- Next 30 days
- Next 60 days

Content summary

Show:

- Researching
- Draft
- Ready for Design
- Designing
- Ready for Review
- Correction Required
- Ready to Publish
- Published

Recent activity

Show recent:

- scholarship updates
- professor additions
- university updates
- opportunity additions
- content updates
- emergency uploads

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7. SHARED COUNTRY SYSTEM

Create one reusable `countries` table.

Fields:

- id
- country_name
- country_code
- flag/icon optional
- active
- display_order
- created_at

- updated_at

Admin can:

- Add country
- Edit country
- Activate/deactivate
- Reorder

Do NOT hard-code a maximum number of countries.

The same country system can be used by:

- Scholarships
- Universities
- Professors
- Co-Curricular
- Content

However, each module must maintain its own appropriate relationships.

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8. SCHOLARSHIPS

Create a complete scholarship research database.

Main table:

`scholarships`

Core fields:

- Scholarship Name
- Provider
- Country
- Degree Level
- Funding
- Opening Date
- Closing Date
- Status
- Assigned To
- Verified By
- Last Updated

- Official Link
- Application Link
- Notes

Additional information:

Eligibility

- Nationality
- Bangladesh Eligible
- International Students Eligible
- Minimum GPA
- IELTS
- TOEFL
- SAT
- GRE
- GMAT
- TOPIK
- Age Requirement
- Academic Background
- Work Experience
- Financial Need
- Other Eligibility
- Eligibility Summary

Funding

Support multiple funding types:

- Fully Funded
- Fully Funded / Partial
- Partially Funded
- Tuition Waiver
- Tuition Reduction
- Stipend
- Living Allowance
- Travel Grant
- Accommodation
- Insurance
- Application Fee Waiver
- Research Funding
- Other
- Self Funded

Funding details:

- Tuition Coverage
- Tuition Percentage
- Monthly Stipend
- Annual Stipend
- Accommodation
- Airfare
- Insurance
- Visa Support
- Research Grant
- Meal Allowance
- Funding Duration
- Funding Notes

Documents

Support:

- Passport
- National ID
- Birth Certificate
- Transcript
- Academic Certificate
- Recommendation Letter
- SOP
- Personal Statement
- CV
- Study Plan
- Research Proposal
- Motivation Letter
- Financial Documents
- Language Certificate
- Portfolio
- Medical Certificate
- Other

Application

- Application Method
- Application Portal
- Application URL
- Application Fee
- Currency
- Fee Waiver
- Separate University Application

- Separate Scholarship Application
- Nomination Required
- Professor Contact Required
- Professor Acceptance Required
- Recommendation Required
- Application Steps
- Application Notes

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9. SCHOLARSHIP STATUS

Do not use one generic status field for everything.

Separate:

Research Status

- New
- Researching
- Fact Checking
- Verified
- Needs Correction
- Rejected

Publication Status

- Not Ready
- Preparing
- Ready for Review
- Approved
- Published
- Unpublished

Deadline State

Automatically calculate:

- Upcoming
- Closing Soon
- Closed

Show:

- Deadline
- Days Remaining

Do not manually enter days remaining.

---

10. UNIVERSITIES

The University module has its own hierarchy:

Country
→ University
→ Program
→ Intake / Deadline / Requirements

University module is separate from Professor module.

Country cards should be dynamically created.

Example:

USA
Qatar
South Korea
Italy
Hungary

But do not limit countries.

Click a country → show universities.

University fields:

- University Name
- Country
- City
- University Type
- Official Website
- Application Status
- Assigned To
- Verified
- Last Updated
- Acceptance Rate
- Description

- Notes

University Type:

- Public
- Private
- National University
- Research University
- Technical University
- University of Applied Sciences
- Other

---

11. UNIVERSITY PROGRAMS

Click University → show programs.

Program fields:

- Program Name
- Department
- Faculty
- Degree Level
- Field of Study
- Language
- Program Website
- Application URL
- Application Status
- Tuition Fee
- Currency
- Tuition Period
- Application Fee
- Application Fee Currency
- Academic Requirements
- Admission Requirements
- Eligibility
- Bangladesh Eligible
- Scholarship Available
- Notes
- Verified

---

12. UNIVERSITY INTAKES

Support:

- Spring
- Summer
- Fall
- Winter
- Rolling
- Multiple Intakes
- Other

Fields:

- Intake Year
- Opening Date
- Closing Date
- Status
- Source URL
- Verified

---

13. UNIVERSITY RANKINGS

Ranking MUST be included.

Fields:

- Ranking Organization
- Ranking Year
- Overall Rank
- Regional Rank
- Subject Rank
- Source URL
- Verified
- Last Updated

Never invent ranking data.

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14. PROFESSOR MODULE — IMPORTANT

This module MUST be independent from the University module.

The hierarchy is:

Country

↓

University

↓

Department

↓

Professor

The user should be able to create these records manually one by one.

14.1 PROFESSOR COUNTRY

When clicking Professors:

Show country cards/list.

Example:

- USA
- Canada
- South Korea
- Japan
- Mexico

- Russia
- Qatar
- Italy
- Hungary

But users/admins can add countries dynamically.

No artificial limit.

14.2 PROFESSOR UNIVERSITY

Click a country.

Show universities under that country.

The user can:

- Add University
- Edit University
- Search University
- Delete/archive if permitted

University fields:

- University Name
- Country
- Official University Link
- City optional
- Notes
- Assigned To
- Verified By
- Last Updated

IMPORTANT:

This university record belongs ONLY to the Professor Research system.

It does NOT automatically create or modify a university in the main University section.

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15. PROFESSOR DEPARTMENT

Click a professor-research university.

Show:

Departments

Users can add departments one by one.

Example:

- Computer Science
- Artificial Intelligence
- Electrical Engineering
- Mechanical Engineering
- Information Technology

Fields:

- Department Name
- Department Official Link
- Research Areas
- Description
- Assigned To
- Verified By
- Last Updated
- Notes

There must be NO artificial limit on departments.

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16. PROFESSOR RECORD

Click a department.

Show professor list.

Professor fields MUST include:

- Professor Name
- Professional Email
- Designation
- Recent Research

- Previous Research
- Research Areas
- Official Profile Link
- Lab Name
- Lab Link
- Assigned To
- Verified By
- Last Updated
- Notes

Optional:

- Publications
- Google Scholar URL
- ORCID
- Personal Website

Do not invent information.

If information is unknown:

- Unknown
- Not publicly available
- Not verified

---

17. PROFESSOR TABLE

Use an Excel/Airtable-style table.

Columns:

- Serial
- Professor Name
- Designation
- Research Areas
- Recent Research
- Previous Research
- Professional Email
- Official Profile
- Assigned To
- Verified By
- Last Updated

Features:

- Search
- Sort
- Filter
- Inline edit
- Pagination
- Column visibility
- Column resizing
- Row selection
- Bulk update
- CSV import/export

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18. PROFESSOR IMPORT

Support:

- CSV
- Excel-compatible CSV
- JSON
- Structured text

Workflow:

Upload/Paste
→ Parse
→ Map Fields
→ Validate
→ Preview
→ Duplicate Detection
→ Compare
→ Confirm
→ Save

Possible duplicate professor matching:

- Name
- University
- Department
- Email
- Official Profile

Never automatically overwrite verified information.

---

19. CO-CURRICULAR

The Co-Curricular section must contain EXACTLY these four main subsections:

1. Leadership

2. Fellowship

3. Internship

4. Emergency Upload / Recent News Update

Use attractive cards for these four sections.

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20. LEADERSHIP

Fields:

- Opportunity Name
- Organization
- Country
- City
- Opening Date
- Closing Date
- Eligibility
- Requirements
- Funding

- Official Link
- Application Link
- Assigned To
- Verified By
- Research Status
- Publication Status
- Last Updated
- Notes

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21. FELLOWSHIP

Fields:

- Fellowship Name
- Organization
- Country
- City
- Duration
- Opening Date
- Closing Date
- Eligibility
- Requirements
- Funding Amount
- Tuition Coverage
- Stipend
- Travel
- Accommodation
- Official Link
- Application Link
- Assigned To
- Verified By
- Research Status
- Publication Status
- Last Updated
- Notes

---

22. INTERNSHIP

Fields:

- Internship Name
- Organization
- Country
- City
- Opening Date
- Closing Date
- Requirements
- Eligibility
- Funding
- Official Link
- Application Link
- Assigned To
- Verified By
- Research Status
- Publication Status
- Last Updated
- Notes

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23. EMERGENCY UPLOAD / RECENT NEWS UPDATE

This is extremely important.

Create a dedicated subsection called:

Emergency Upload

or

Recent News Update

This section is for information that needs to be uploaded/published urgently.

Examples:

- New scholarship announcement
- Deadline extended
- New application opened
- Important university announcement
- Funding update
- Emergency correction
- New opportunity
- Important breaking information

Fields:

- Title
- Information Type
- Country
- Related Scholarship
- Related University
- Related Opportunity
- Source URL
- Source Name
- Information
- Upload Date
- Urgency
- Assigned To
- Verified By
- Status
- Content Status
- Notes

Urgency:

- Normal
- High
- Emergency

Emergency records should be highly visible to the Content Manager.

---

24. CONTENT MANAGEMENT — MOST IMPORTANT WORKSPACE

The Content section should be designed primarily for the **Content Manager**.

The Content Manager should be able to understand everything important from ONE dashboard.

Do not make the Content Manager repeatedly open:

- Scholarships
- Co-Curricular
- Deadlines
- Emergency Upload

just to find what needs publishing.

The Content Dashboard should aggregate the relevant information automatically.

---

25. CONTENT DASHBOARD LAYOUT

Create the Content Dashboard in this priority order:

TOP PRIORITY — EMERGENCY UPLOAD

At the very top show a prominent section:

🚨 EMERGENCY UPLOAD

Show:

- New emergency information
- Recent news updates
- Urgent scholarship updates
- Deadline changes
- Important university updates
- Newly discovered opportunities

Each item should show:

- Title
- Type
- Source
- Date
- Urgency
- Assigned To

- Verification status
- Content status

Provide a clear action:

Create Content

or

Process Now

This section MUST be visually more prominent than normal content.

---

26. UPCOMING DEADLINES

Below Emergency Upload, show:

Upcoming Deadlines

Automatically collect deadlines from:

- Scholarships
- University applications
- Programs
- Leadership
- Fellowships
- Internships

Display buckets:

Next 5 Days

Next 7 Days

Next 30 Days

Next 60 Days

Each row:

- Name
- Type
- Country
- University/Organization
- Deadline
- Days Remaining
- Application Status
- Content Status

Clicking a row should open the source record.

Do NOT duplicate deadline data inside Content.

Use relational references.

---

27. APPLICATION START INFORMATION

The Content Manager must be able to see when applications are starting.

Create a section:

Application Starting Soon

Show:

- Scholarship
- University
- Program
- Opportunity
- Country
- Application Opening Date
- Closing Date
- Days Until Opening
- Content Status

Sort by nearest opening date.

---

28. READY TO PUBLISH

Create a highly visible section:

Ready to Publish

Show content that has completed the required workflow.

Fields:

- Content Title
- Related Record
- Type
- Country
- Assigned Designer
- Editor
- Status
- Ready Date
- Notes

Provide action:

Publish

Only authorized users can publish.

---

29. PUBLISHED CONTENT

Create:

Published

Show:

- Content Title
- Type
- Related Record
- Country
- Published Date
- Published By
- Designer
- Editor
- Public URL
- Notes

Allow filtering by:

- Date
- Type
- Country
- Related module
- Publisher

---

30. NEWLY UPLOADED INFORMATION

Create a section:

New Information

Automatically show newly created or updated records from:

- Scholarships
- Co-Curricular
- Emergency Upload
- Universities where relevant
- Other approved content sources

Show:

- Title
- Type
- Created/Updated Date
- Updated By
- Verification Status

- Content Status

This helps the Content Manager discover new information without manually checking every module.

---

31. CONTENT WORKFLOW

Use a structured workflow:

Idea
→ Researching
→ Draft
→ Ready for Design
→ Designing
→ Ready for Review
→ Correction Required
→ Ready to Publish
→ Published
→ Archived / Unpublished

Do not automatically publish content simply because it reaches a status.

---

32. CONTENT RECORD

`contents`

Fields:

- id
- title
- content_type
- related_entity_type
- related_entity_id
- country_id
- content_status
- assigned_researcher
- designer
- editor

- publisher
- planned_date
- published_date
- public_url
- notes
- created_at
- updated_at

Content should reference the original source records.

Do NOT copy the entire scholarship/opportunity data into Content.

---

33. CONTENT SEARCH

Content Manager must have a powerful search.

Search by:

- Title
- Scholarship
- University
- Program
- Leadership
- Fellowship
- Internship
- Emergency Upload
- Country
- Organization
- Status

Use debounced search.

---

34. CENTRAL DEADLINES

Create a central deadline architecture.

Deadlines can come from:

- Scholarships

- Universities
- Programs
- Leadership
- Fellowship
- Internship
- Content planned dates

Support:

- Opening Date
- Closing Date
- Deadline Type
- Time
- Timezone
- Source URL
- Verification
- Notes

Prepare architecture for future Google Calendar integration.

Do not require Google OAuth in the first implementation.

---

35. VERIFICATION CENTER

Create a central verification dashboard.

Show records requiring verification from:

- Scholarships
- Universities
- Programs
- Professors
- Leadership
- Fellowship
- Internship
- Emergency Upload

Show:

- Record
- Type
- Assigned Researcher

- Verifier
- Missing Information
- Verification Progress
- Last Updated

Click → open detailed verification checklist.

---

36. ACTIVITY LOG

Create immutable `activity_logs`.

Fields:

- user_id
- action
- entity_type
- entity_id
- field_name
- old_value
- new_value
- timestamp

Track:

- Created
- Edited
- Assigned
- Verified
- Rejected
- Published
- Unpublished
- Imported
- Deleted where permitted

Example:

“User changed application deadline from 15 March to 30 March.”

Normal users cannot edit historical activity logs.

---

37. DATA QUALITY

Every major record should have a completeness score.

Example:

- Scholarship — 92%
- University — 86%
- Professor — 95%

These are examples only.

Calculate automatically based on:

- required fields
- verification status
- source availability

Show:

- completeness percentage
- missing fields
- verification progress

Optional fields should not unfairly reduce completeness.

---

38. DUPLICATE DETECTION

Before creating/importing records, detect possible duplicates.

Scholarship:

- Name
- Provider
- Country
- Official URL

University:

- Name
- Country
- Official Website

Professor:

- Name
- University
- Department
- Email
- Official Profile

Opportunity:

- Name
- Organization
- Country
- Official URL

Show:

Possible Duplicate

before saving.

User can:

- Merge
- Keep Existing
- Create Anyway
- Cancel

---

39. EXCEL-LIKE EXPERIENCE

Major tables must support:

- Search
- Sort
- Multiple filters
- Pagination
- Inline editing
- Dropdown
- Multi-select
- Date picker

- Checkbox
- Column resizing
- Column hiding
- Row selection
- Bulk update
- CSV export

Creation should primarily happen through structured forms.

Small changes can happen inline.

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40. FILTERING

All major sections must support multiple filters simultaneously.

Examples:

Scholarships:

Bachelor
+
South Korea
+
Fully Funded
+
Upcoming

Professor:

South Korea
+
Computer Science
+
AI Research

Co-Curricular:

Internship
+
South Korea
+
Open

Content:

Ready to Publish
+
South Korea
+
Scholarship

---

41. SAVED FILTERS

Users can save frequently used filters.

Examples:

- Fully Funded Undergraduate
- South Korea Opportunities
- Upcoming 7-Day Deadlines
- Ready to Publish
- Emergency Information
- AI Professors

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42. IMPORT / EXPORT

Support CSV import/export.

Export:

- All records
- Filtered records
- Selected records

Import workflow:

Upload
→ Parse
→ Field Mapping
→ Validation
→ Preview
→ Duplicate Detection

→ Comparison
→ Confirmation
→ Save

Never blindly overwrite verified information.

---

43. SOURCE VALIDATION

Official sources must be preferred.

Never invent:

- deadlines
- tuition
- ranking
- acceptance rate
- eligibility
- funding
- requirements
- application links
- professor emails
- research information

If unknown:

- Unknown
- Not publicly available
- Not verified

---

44. SEARCH ARCHITECTURE

Each major section must have a visible search box.

Search should support:

Scholarships
Universities
Professors
Departments

Co-Curricular
Content
Deadlines
Verification

Use indexed database queries.

For large datasets use:

- server-side search
- debouncing
- pagination
- appropriate indexes

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45. PERFORMANCE

Design for:

- thousands of scholarships
- thousands of universities
- thousands of professors
- tens of thousands of programs
- tens of thousands of content records

Use:

- server-side pagination
- database indexes
- efficient joins
- debounced search
- lazy loading
- virtualized tables where appropriate
- selective data fetching

Never load the entire database into the browser.

---

46. DATABASE ARCHITECTURE

Core tables:

Authentication

- profiles
- roles
- permissions if necessary

Shared

- countries
- degree_types
- funding_types
- statuses
- fields_of_study

Scholarships

- scholarships
- scholarship_degrees
- scholarship_funding
- scholarship_deadlines
- scholarship_documents
- scholarship_sources
- scholarship_assignments
- scholarship_verification_items

Main Universities

- universities
- university_programs
- program_intakes
- university_deadlines
- university_documents
- university_rankings
- university_sources
- university_scholarships

Professor Research System

Use separate professor-research hierarchy:

- professor_countries or shared countries with module relationship
- professor_universities

- departments
- professors
- professor_publications
- professor_sources
- professor_verification_items

IMPORTANT:

`professor_universities` must NOT depend on the main `universities` table.

This is intentional.

Co-Curricular

- opportunities
- opportunity_sources
- opportunity_deadlines
- opportunity_assignments
- opportunity_verification_items

Use an opportunity type field:

- Leadership
- Fellowship
- Internship
- Emergency Upload

Content

- contents
- content_assignments
- content_history

System

- activity_logs
- saved_filters
- daily_notes
- notifications
- settings

---

47. DATABASE RELATIONSHIPS

Main University system:

Country
→ University
→ Program
→ Intake
→ Deadline

Scholarship:

Country
→ Scholarship
→ Funding
→ Deadline

Professor system:

Country
→ Professor University
→ Department
→ Professor

Co-Curricular:

Country
→ Opportunity

Content:

Content
→ Related Scholarship / Related Co-Curricular / Related University / Related information

Users:

User
→ Assignment
→ Verification
→ Activity

Do not create unnecessary duplicated entities.

---

48. SECURITY

Use Supabase Row Level Security.

Never expose:

- service-role key
- internal database credentials
- private user information

Protect:

- team data
- internal notes
- verification notes
- activity logs
- private research information

No public access to internal data.

No public registration.

No search engine indexing requirement.

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49. PUBLIC SCHOLAR COMPATIBILITY

The internal system may eventually feed the public Scholar platform.

Only explicitly approved/published information can become public.

Never expose publicly:

- assigned researcher
- internal notes
- verification notes
- activity logs
- private user information
- internal workflow information

Keep public-facing data logically separated from internal data.

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50. SETTINGS

Admin should be able to manage:

- Countries
- Degree Levels
- Funding Types
- Research Areas
- Opportunity Types
- Statuses
- Team Members
- Roles
- Verification Requirements
- Publication Requirements
- Deadline Warning Thresholds

Do not unnecessarily hard-code configurable values.

---

51. NOTIFICATIONS

Prepare architecture for:

- Assignment
- Verification Request
- Correction Request
- Verification Complete
- Deadline Approaching
- Content Assigned
- Designer Assigned
- Ready to Publish
- Emergency Upload

Start with in-app notifications.

Email notifications can be added later.

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52. RESPONSIVE DESIGN

Desktop-first.

Also support:

- Laptop
- Tablet
- Mobile

On mobile:

- tables become cards or horizontally scrollable
- forms stack vertically
- search remains accessible
- country cards remain usable
- important Content Manager alerts remain visible

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53. UI PRIORITY

The application should feel like an actual internal operating system.

Do NOT make every section look like a separate website.

Use shared:

- sidebar
- topbar
- cards
- tables
- filters
- search
- modals
- drawers
- badges
- forms
- pagination
- empty states
- loading states
- error states

Maintain visual consistency across the entire application.

---

54. DEVELOPMENT ORDER

Implement in this order:

STEP 1 — Inspect existing repository

STEP 2 — Project architecture

STEP 3 — Supabase connection

STEP 4 — Authentication and email invitation

STEP 5 — Profiles, roles and RLS

STEP 6 — Shared country/configuration system

STEP 7 — Main layout/sidebar

STEP 8 — Dashboard

STEP 9 — Scholarship database

STEP 10 — Scholarship forms/editing

STEP 11 — Scholarship deadlines/verification

STEP 12 — Main University database

STEP 13 — Programs/intakes/deadlines/rankings

STEP 14 — Independent Professor Research system

STEP 15 — Professor countries

STEP 16 — Professor universities

STEP 17 — Professor departments

STEP 18 — Professor database

STEP 19 — Professor import/verification

STEP 20 — Co-Curricular

STEP 21 — Leadership

STEP 22 — Fellowship

STEP 23 — Internship

STEP 24 — Emergency Upload / Recent News

STEP 25 — Content Manager Dashboard

STEP 26 — Content workflow

STEP 27 — Deadline intelligence

STEP 28 — Verification Center

STEP 29 — Activity Log

STEP 30 — Search/filter/saved filters

STEP 31 — Import/export

STEP 32 — Duplicate detection

STEP 33 — Data quality

STEP 34 — Performance optimization

STEP 35 — Security/RLS audit

STEP 36 — Full end-to-end testing

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55. CRITICAL ACCEPTANCE TEST

Authentication

1. Admin invites user
2. User receives email
3. User authenticates
4. User enters application
5. Unauthorized user cannot enter
6. Logout works

Scholarships

7. Create scholarship

8. Edit scholarship
9. Search scholarship
10. Filter scholarship
11. Assign researcher
12. Verify scholarship
13. Add deadline
14. View deadline
15. Mark ready to publish
16. Publish

Universities

17. Create country
18. Open country
19. Add university
20. Add program
21. Add intake
22. Add deadline
23. Add ranking
24. Add requirements
25. Add tuition
26. Add application fee
27. Add scholarship
28. Search university
29. Inline edit

Professors

30. Open Professor section
31. Add country
32. Add professor-research university
33. Add department
34. Add professor
35. Add professional email
36. Add designation
37. Add recent research
38. Add previous research
39. Assign user
40. Verify professor
41. Search professor
42. Import multiple professors

IMPORTANT:

Professor university records must remain independent from the main University module.

Co-Curricular

43. Open Leadership
44. Add Leadership opportunity
45. Open Fellowship
46. Add Fellowship
47. Open Internship
48. Add Internship
49. Open Emergency Upload
50. Add emergency information
51. Search
52. Filter
53. Verify

Content Manager

54. Open Content Dashboard
55. See Emergency Upload at top
56. See upcoming deadlines
57. See application starting soon
58. See newly uploaded information
59. See Ready to Publish
60. See Published Content
61. Create content
62. Assign content
63. Move content through workflow
64. Publish content
65. Search content
66. Filter content

Daily Work

67. Add daily note
68. Record work
69. View weekly productivity
70. View monthly productivity

System

71. Activity log records changes

72. Duplicate detection works
73. Import validation works
74. RLS prevents unauthorized access
75. Large dataset search works
76. No artificial record/block limits exist
77. Existing functionality remains intact

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56. FINAL ARCHITECTURAL REQUIREMENT

The final application must function as one unified:

SCHOLAR INTERNAL RESEARCH OPERATING SYSTEM

Dashboard

Scholarships

→ Country

→ Scholarship

→ Provider

→ Funding

→ Eligibility

→ Deadline

→ Documents

→ Application

→ Official Sources / Links

→ Verification

→ Publication

→ Notes

→ Custom Fields

Universities

→ Country

→ University

→ State / Region

→ Program

→ Intake

→ Deadline

→ Tuition

→ Application Fee

→ Requirements

→ Ranking

→ Scholarships

→ Official Sources / Links

→ Verification

→ Notes

→ Custom Fields

Professors

→ Country

→ Professor University

→ Department

→ Professor

→ Research

→ Email

→ Profile

→ Lab / Website

→ Assignment

→ Verification

→ Official Sources / Links

→ Notes

→ Custom Fields

Co-Curricular

→ Leadership

→ Fellowship

→ Internship

→ Emergency Upload / Recent News

→ Notes

→ Verification

→ Official Sources / Links

Content

→ Emergency Information

→ New Information

→ Upcoming Deadlines

→ Application Starting Soon

→ Research

→ Draft

→ Design

→ Review

→ Ready to Publish

→ Published

→ Archived / Unpublished

Deadlines

→ All Relevant Deadlines

→ Scholarships

→ Universities / Programs

→ Co-Curricular

→ Content

→ Calendar / Timeline

Verification

→ All Records Requiring Verification

→ Scholarship Verification

→ University Verification

→ Professor Verification

→ Co-Curricular Verification

→ Content Verification

Team

→ Users

→ Roles

→ Assignments

→ Responsibilities

→ Workload

Activity Log

→ Complete History

→ Record Changes

→ Verification History

→ Assignment History

→ Publication History

Settings

→ Countries

→ Categories

→ Degree Levels

→ Funding Types

→ Statuses

→ Verification Configuration

→ Publication Configuration

→ Custom Fields / Columns

→ Team & Roles

→ System Configuration

MOST IMPORTANT DESIGN PRINCIPLE

The **Content Manager should have a single powerful dashboard where the most important information is surfaced automatically.**

The priority on the Content Dashboard should be:

1. 🚨 Emergency Upload / Recent News
2. Upcoming Deadlines
3. Application Starting Soon
4. Newly Uploaded Information
5. Ready to Publish
6. Published Content
7. Content Productivity

The Content Manager should not need to manually inspect every other module to understand what requires action.

The system should automatically pull relevant information from Scholarships and Co-Curricular and present it in the Content workspace.

FINAL DEVELOPMENT RULE

Before implementing anything:

1. Inspect the existing repository.
2. Understand existing architecture.
3. Preserve working functionality.
4. Identify reusable components.
5. Identify existing database tables.
6. Avoid duplicate tables where existing structures can be reused safely.

7. Implement the database architecture first.
8. Implement RLS and authentication securely.
9. Build reusable UI components.
10. Build modules according to the development order above.
11. Test each module before moving to the next.
12. Do not mark the application complete until the acceptance tests pass.

Build a real, scalable, database-backed internal platform.

Do NOT build a static prototype.

Do NOT create disconnected mini-applications.

Do NOT create artificial record limits.

Do NOT invent research, deadlines, rankings, tuition, eligibility, funding, emails, or other factual data.

The final product should feel like a professional internal research operating system used by a real scholarship research and content team.

XX. FLEXIBLE COLUMN / CUSTOM FIELD SYSTEM — GLOBAL REQUIREMENT

This requirement applies to all major research databases:

- Scholarships
- Universities
- Programs
- Professors
- Leadership
- Fellowship
- Internship
- Emergency Upload / Recent News
- Content
- Other future research entities

The application must use a flexible database-table architecture.

XX.1 DEFAULT + CUSTOM FIELDS

Every module should have:

1. Developer-defined core fields
2. User-defined custom fields

Core fields must remain stable and structured.

Custom fields allow researchers/admins to extend the database without developer intervention.

XX.2 ADD COLUMN

Every major table should have:

- Add Column

Authorized users can create a new field by specifying:

- Column Name
- Type
- Description
- Required / Optional
- Default Value
- Visibility
- Editable / Read Only

XX.3 SUPPORTED FIELD TYPES

Support at minimum:

- Text
- Long Text
- Number
- Currency
- Percentage
- Date
- Date & Time
- Checkbox
- Single Select
- Multi Select
- URL
- Email
- Rating
- User
- Status

Architecture should remain extensible for future field types.

XX.4 TABLE CUSTOMIZATION

Users must be able to:

- Add columns
- Rename columns
- Reorder columns
- Resize columns
- Hide columns
- Show hidden columns
- Freeze columns
- Sort columns
- Filter columns
- Search across columns
- Create saved views
- Delete/archive custom columns where permitted

XX.5 NOTES FIELD

Every major entity must support notes.

Notes should not be restricted to a short one-line field.

Support:

- General Notes
- Internal Notes
- Research Notes
- Important Notes
- Additional Comments

XX.6 OFFICIAL SOURCE FIELDS

Research-oriented entities must support official source URLs.

Users should be able to add multiple source links where appropriate.

Each source should have:

- Source Name
- Source Type
- URL
- Description
- Verified
- Last Checked

XX.7 MANUAL CHECKING

Research entities should support configurable manual verification checklists.

Admin must be able to define checklist items.

Each checklist item supports:

- Checked
- Not Checked
- Needs Correction
- Not Applicable

The system calculates verification progress automatically.

XX.8 DATABASE ARCHITECTURE

Custom fields must be database-backed.

Do NOT create a frontend-only solution.

Recommended architecture:

`custom_field_definitions`

- id
- entity_type
- field_name
- field_type
- description
- required
- default_value
- options
- display_order
- active

- created_by
- created_at
- updated_at

`custom_field_values`

- id
- entity_type
- entity_id
- field_definition_id
- value
- updated_by
- updated_at

This architecture must allow the team to extend tables without requiring code changes for every new research field.

XX.9 SECURITY

Only authorized roles can create, edit or delete custom field definitions.

Suggested permissions:

Admin:
Full control

Researcher:
Use existing custom fields; create fields if permitted

Verifier:
Edit verification-related fields

Content Manager:
Use content-related fields

Viewer:
Read only

Every structural change must be recorded in Activity Log.

XX.10 NO ARTIFICIAL LIMITS

Do not introduce arbitrary limits such as:

- Maximum 20 columns
- Maximum 100 records
- Maximum 50 custom fields
- Maximum 10 countries

The system should scale through:

- Pagination
- Server-side filtering
- Indexed search
- Virtualized tables
- Lazy loading
- Efficient database queries
- Appropriate indexing